

Fuzzy Logic-based Adaptive Traffic Light Control of an Intersection: A Case Study

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Abstract— The increasing population in the world causes heavy traffic conditions, especially in metropolises. The traffic problem is the main challenge for the city. Considering the reasons for the traffic density in Turkey, the traffic lights schedule is shown in the first among when comparing with other parameters. Determining the green light durations according to the vehicle density will reduce the average waiting times of vehicles at intersections. This study performed different traffic scenarios based on VANET on SUMO for adaptive and non-adaptive intersections. The gathered traffic information data from vehicles are given to the developed fuzzy logic model to optimize green light durations. A Period of a scenario for analyzing took 10 minutes, and according to 10 minutes input, the fuzzy model optimizes the green light durations for the following period. Test results show that using a fuzzy model in traffic light optimization decreases the average waiting time of vehicles and average queue length.

Keywords— *sumo, vanet, fuzzy logic, traffic lights, smart city*

I. INTRODUCTION

One of the main challenges for developed cities is vehicle traffic and its management. Because of physical structures and crossroads of urban intersections, traffic flow may cause time delay, congestion, traffic accidents, and more fuel/time consumption [1]. To reduce or prevent these challenges in traffic, signaling systems are used in urban intersections. These signaling systems are generally divided into four main groups; isolated systems that affect only one intersection, coordinate systems that affect two intersections together, progressive systems that work according to pre-set rules, and finally, synchronous systems that coordinate the same signal phase for all intersections [2-3].

Nowadays, these traffic signaling systems that have been subject to academic and commercial studies are used in many cities worldwide. For example, Transty [4] and Scats [5] are the most known systems that manage traffic signalization according to different time zones and traffic congestion. Unlike Transty and Scats, Rhodes [6] and Scoot [4] use signal loops to manage traffic.

But these mentioned methods have weaknesses such as lack of flexibility, programmability, performance, and especially non – real-time criteria in their structure. These systems have become insufficient for managing traffic at urban intersections. Therefore, unlike existing traffic signalization systems, there are many different studies about intelligent computational techniques to solve traffic signaling problems such as fuzzy logic, artificial neural

networks, genetic algorithm, etc. [7-12]. For example, in the traffic light optimization study [7], a performance improvement of 10.13% was obtained using particle swarm optimization. In another study similar to ours, the authors proposed a fuzzy-based model and got 8% performance in average vehicle speed [10]. Ahmad et al. define the earliest deadline first (EDF) algorithm to reduce traffic congestion in urban and succeed in reducing the average waiting time [11]. Also, studies can be found on vehicle-to-vehicle communication to reduce traffic. For example, a study suggested a model that would provide convenience to drivers, especially in stop-and-go traffic [13].

Researchers try to solve the most common problems in traffic signaling using computational techniques, and important ones are listed below.

- Cycle time optimization
- Green time optimization
- Queuing size optimization
- Average vehicle speed optimization
- Routing/information control
- Prediction of traffic density in an intersection

In this paper, greenlight duration optimization was performed, which is the main problem of adaptive traffic management systems. In this study, data used in the model was obtained from VANET architecture (Vehicular Ad Hoc Networks) via DSRC (Dedicated short-range communications). A case study according to Turkey's traffic signaling system that allows only one –way right at intersections has been performed on different VANET scenarios using SUMO and new green light durations obtained from the proposed fuzzy logic model was compared each other.

The rest of the paper is organized as follows: information about traffic scenarios and frameworks is provided in the following section. The proposed fuzzy logic model will be explained in Section III. The study's simulations and results will be analyzed in Section IV, while conclusions are presented in the last section.

II. TRAFFIC SYSTEM & FRAMEWORK

A. Traffic System in Turkey

Drivers in Turkey drive on the right side. Traffic lights are heavily used, and red light periods are usually long. The light synchronization principle at the intersections is to burn green light for only one crossroads each time at the

intersection. Since this light system is prevalent in Turkey, our work has been studied on this kind of intersection as a case study. Simulations are made on adaptive and non-adaptive intersections in the study for the metropolitan cities such as Istanbul, Ankara, and Izmir have more than one adaptive intersection.

B. Simulation of Urban Mobility (SUMO)

SUMO is open-source simulation software that allows modeling traffic systems containing objects like public transport vehicles, pedestrians, vehicles, etc., and running different VANET scenarios on them. SUMO is the preferred VANET simulation program in the literature, providing additional features such as pathfinding, visualization, and algorithm development [14-15].

C. Vehicular Ad Hoc Networks (VANET)

VANET is a technology that accepts intelligent vehicle as a node and uses these nodes to build a mobile network. In vehicular networks, communication between vehicles is provided via OBUs (onboard units) embedded in the vehicle. OBUs have 802.11p antennas that use the DSRC (Dedicated Short Range Communications) standard created by modifying the frequency and power parameters of 802.11a. The DSRC standard has a bandwidth of 75 MHz in the 5.9 GHz band and consists of 7 channels each of 10 Mhz. Average data rates on channels are between 6-27 Mbps [16]. The vehicles broadcast the obtained data, such as position, speed, direction, neighborhood table, etc., to other vehicles (V2V) or roadside units (RSU-V2I) via beacon signals.

D. Traffic Simulation Models

In order to be compatible with the traffic system of Turkey, two crossing intersection models like Fig. 1 were created, and two different traffic flows were studied. In the first model, 75% of the total traffic was set from west to east, and the rest of the traffic was set as random. In the second model, 75% of the total traffic was set from south to north, and the rest of the traffic was set again as random. The first model aims to affect the intersections to each other and to create an adaptive intersection. The second model aims to generate non-adaptive intersection traffic, which does not affect each other very much. In both models, only one road in the intersection is green. Vehicles on the green phase can pass freely on other roads in the intersection.

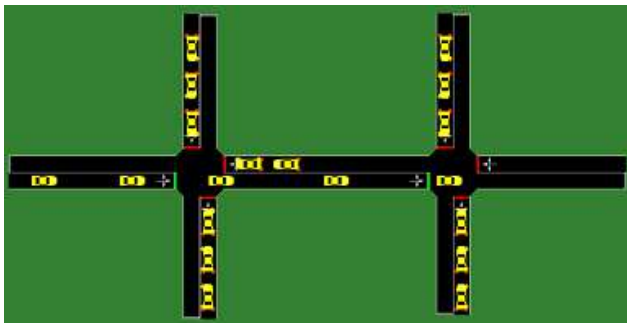


Fig. 1. Intersection model designed in this study

III. PROPOSED METHOD

In this study, each scenario was considered independently, and the optimal green light duration was tried to be estimated using a developed fuzzy logic model. The proposed fuzzy model was used according to the data obtained from vehicles and RSUs (Road Side Units) in VANET traffic scenarios for 10 minutes. After analyzing the results, the new green light duration was adjusted for the following ten minutes-period. Thus, we aimed to reduce the average waiting times of vehicles and average queue length at intersections.

A. Fuzzy Logic

The uncertainty in a system or a concept is called fuzziness. Fuzzy logic, which produces effective results according to uncertain and unclear information, efficiently implements control models [17]. Fuzziness occurs when the general set equation allows partial membership. The general set equation is explained in Eq. 1, and the fuzzy set equation is described in Eq. 2.

$$\mu_A(x): X \rightarrow \{0,1\}$$

$$\mu_A(x) = \begin{cases} 1 & x \in X \\ 0 & x \notin X \end{cases} \quad (1)$$

$$\tilde{A} = \{(x, \mu_{\tilde{A}}(x)) \mid \forall x \in X\} \quad (2)$$

$$\mu_{\tilde{A}}(x): X \rightarrow [0,1]$$

The developed fuzzy logic model takes three inputs and produces optimized green time duration as output for each road at an intersection. In Table 1, there are explanations about the inputs and outputs of the proposed system. The proposed fuzzy model uses 27 rules to determine the green light duration according to Mamdani model.

TABLE 1. Definition of inputs and output of fuzzy model

Abbreviation	Parameter Name	Type
RP	Road Priority	input
AQL	Average Queue Length	input
AWT	Average Waiting Time	input
GT	Green Time	output

The RP from the inputs is the path priority ratio. Although the roads at intersections have equal traffic density, traffic flow will affect vehicle density over roads. In this study, the roads in the direction of traffic flow were accepted to have 0.3 and others 0.2 priority rate. While AQL parameter is used as input for the average vehicle queue in each road at the intersection, AWT is used as input and stands for the average waiting time of vehicles. The membership functions used for inputs in the fuzzy model are given in Fig. 2. In the fuzzy model, “min” is chosen as the “AND” method, “max” is chosen as the “OR” method, and finally, a centroid is used as the “defuzzification” method. Fuzzy sets for RP are determined, such as {low, med, high} and {short, mid, long} are choose as fuzzy sets for AQL and AWT. The membership function used for the output of the fuzzy model is given in Fig. 3. Fuzzy sets for GT are determined, such as {short, mid, long}.

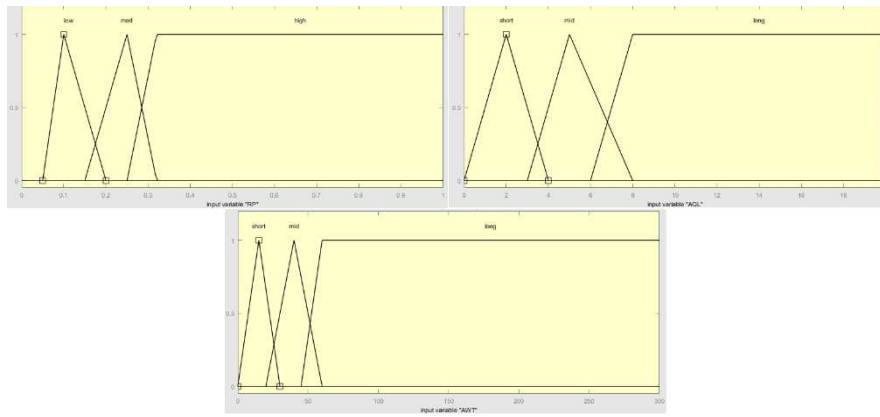


Fig. 2. Membership functions for inputs

For example, to determine the green light duration for road1, average queue length, waiting time, and road priority for the last ten minutes are gathered to compute the model as shown in Eq. 3.

$$\text{road1} = \text{func}([0.3 \ 3 \ 24.3], \text{fuzzymodel}) \quad (3)$$

According to these parameters, the fuzzy model will produce 11.53 seconds for the green light duration, and then this green time will be applied to road1 for the following ten minutes' period.

IV. SIMULATIONS & RESULTS

In this study, two different main scenarios were used. While the first main scenario is applied to adaptive intersections, the second is applied to non-adaptive intersections. These scenarios aim to analyze the improved performance of different types of traffic. In the first main scenario, the traffic density flowing from west to east at the intersections shown in Fig. 1 is set as 75%, and the remaining density is set as random flowing. The total number of vehicles in the system has been increased by 20, starting from 20, and different sub-scenarios have been generated to a maximum of 100 vehicles. For each sub-scenario, the number of vehicles entering the intersection at a specific time interval was increased from 1 to 4, and 4 different sub-studies were applied.

Increasing the number of vehicles per second will increase traffic density to an upper level in every step. For the second scenario, the same number of vehicles and intervals were used. In the second scenario, traffic flow was organized by 75% from south to north and 25% randomly. In the second main scenario, the goal is to create non-adaptive intersections and measure the model's performance at such intersections. At first, all green light durations were set to 20 seconds and yellow light durations to 2 seconds for both scenarios. For the following period, green light durations are determined by the fuzzy model. All yellow lights are taken constant, and they are two seconds in each case.

In the case study, four different studies initiated from 20 vehicles and increased to 100 vehicles were applied for each sub-scenario. The total number of vehicles is kept constant. The number of arriving vehicles per a specific time at intersections has increased in each sub scenarios.

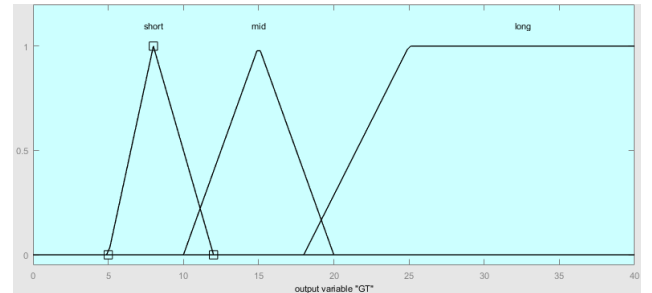


Fig. 3. Intersection model designed in this study

Contents of scenarios and sub-scenarios are given in Table 2. As can be seen from Table 2, 0.71 and 1.43 incoming vehicles per second represent light and medium traffic density, and 2.14 2.86 incoming vehicles per second indicate heavy and very heavy density traffic in the studies.

After all, scenarios have been completed, the results of average waiting time at intersections are given in Fig. 4. Five different scenarios based on two different main scenarios are listed as S1, S2, S3, S4, and S5. While the continuous lines are the results of the first main scenario representing the traffic density flowing from west to east, the discrete lines show the results of the second main scenario, which means the traffic density flowing from south to north. Based on the fuzzy model, the continuous red lines show the average waiting time in the intelligent traffic light system. In contrast, the continuous blue lines show the average waiting times for the constant green light.

The results in Fig. 4 clearly show that the fuzzy-based intelligent traffic light model reduces the average waiting times of vehicles at intersections.

The average waiting time in the fuzzy model is less in each case; as the traffic density increases, the average waiting time in the fuzzy logic model becomes less in all cases. The proposed fuzzy model gives better performance in adaptive intersections, as shown in Fig. 4. According to average general results in Table 3, overall average waiting times for model where the green light is constant and the same for all roads at the intersection are 46.87 and 32.98., these times are reduced to 42.43 in adaptive intersections and 29.81 seconds at non-adaptive intersections in the fuzzy-based model. The reduction performance is better in adaptive intersections.

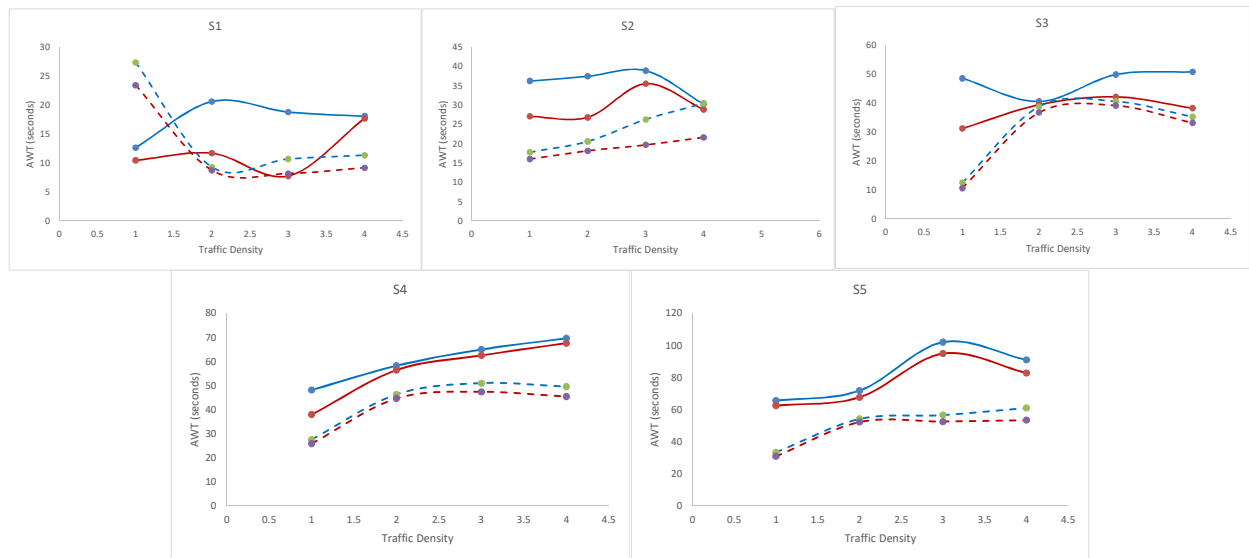


Fig. 4. Average waiting time performances of scenarios

TABLE 2. Name of the scenarios and their definitions

Scenario Name	Sub-Name	#cars per second	Traffic Density
S1	S1_1	0.71	20
	S1_2	1.43	
	S1_3	2.14	
	S1_4	2.86	
S2	S2_1	0.71	40
	S2_2	1.43	
	S2_3	2.14	
	S2_4	2.86	
S3	S3_1	0.71	60
	S3_2	1.43	
	S3_3	2.14	
	S3_4	2.86	
S4	S4_1	0.71	80
	S4_2	1.43	
	S4_3	2.14	
	S4_4	2.86	
S5	S5_1	0.71	100
	S5_2	1.43	
	S5_3	2.14	
	S5_4	2.86	

V. CONCLUSION

Many adaptive intersections are connected in metropolitan cities with high populations. Sometimes these intersections cause long vehicle queues and inexhaustible traffic density in Turkey. In this study, to solve this kind of challenge in traffic, different VANET scenarios run on SUMO and

obtained results related to vehicle (id, speed, route, etc.) are used in the fuzzy model to optimize green light duration. Two different traffic flows are considered, such as west-east and south-north. All scenarios show that the fuzzy-based smart traffic light model has more success than the fixed green light model in average waiting time and average queue length. In addition, the developed fuzzy model has succeeded in reducing average waiting time in adaptive intersections much more than non-adaptive intersections, but for average vehicle queue, the developed model is more successful in non-adaptive intersections. The developed fuzzy model improved 11.53% in average waiting time and 8.27% in average queue length. According to these results, especially in big cities, using fuzzy logic-based models in traffic signaling can decrease vehicles' average waiting time and queue length.

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TABLE 3. Average performance results of scenarios in seconds

Scenarios		S* 1	S* 2	S* 3	S* 4	Average
Main Scenario 1	AQL	3.89	3.83	3.63	3.60	3.74
	Fuzzy AQL	3.69	3.64	3.66	3.30	3.57
	AWT	42.19	45.71	54.88	51.89	48.67
	Fuzzy AWT	33.81	40.36	48.54	47.01	42.43
Main Scenario 2	AQL	3.46	3.53	3.51	3.59	3.52
	Fuzzy AQL	2.80	3.02	3.11	3.44	3.09
	AWT	23.68	33.77	36.99	37.49	32.98
	Fuzzy AWT	21.33	32.06	33.32	35.52	29.81

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